6.1 Lines

Name _____ Period ____ Date _____

Pre-Calculus 6.1 · Student sheet

I can…

- find the inclination of a line from its slope or equation.
- find the angle between two lines.
- find the distance between a point and a line.

Vocabulary · fill in the missing word

Inclination — the _____ θ ($0 \le \theta < \pi$) measured counterclockwise from the positive x-axis to the line

Angle between two lines — the _____ of the angles the lines form, $0 < \theta \le \pi/2$

Part 1 · Slope and angle

Graph each line. Measure the angle from the positive x-axis, counterclockwise.

Line	Angle θ	tan θ	Slope
$y = x$			
$y = 2x$			
$y = (1/2)x$			
$y = -x$			

Part 2 · Notes and examples (fill in each note, then work the example)

Inclination and slope: $m =$ _____

Example 1

Find the inclination of $x + y = 2$.

Example 2

Find the inclination of $2x + 3y = 6$. Round to the nearest tenth of a degree.

Angle between two lines: $\tan\theta =$ _____

Example 3

Find the angle between $2x - y = 4$ and $3x + 4y = 12$. Round to the nearest tenth of a degree.

Example 4

Write the equation of the line through (0, 2) with inclination $\frac{\pi}{6}$.

Distance from (x_1, y_1) to $Ax + By + C = 0$: d = _____

Example 5

Find the distance from (4, 1) to $y = 2x + 1$.

Part 3 · Partner practice

Partner 1

Find the inclination of $y = \sqrt{3}\ x - 1$.

Partner 2

Find the distance from (2, −1) to $3x + 4y - 5 = 0$.